

Commissioned Review: Bacterial Carboxyspermidine Biosynthesis in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Nominal bucket:** KEGG [ppu00330](#) — "Arginine and proline metabolism" (30 primary genes; area: amino_acid_metabolism) **Review focus:** the alternative (carboxyspermidine) route of spermidine biosynthesis and its satisfiability in the target strain.

1. Executive Summary

Pseudomonas putida KT2440 synthesizes the polyamine **spermidine** by the **alternative carboxyspermidine pathway**, not by the classical aminopropyltransferase route used by *E. coli* and by its own genus-mate *P. aeruginosa*. The decisive genomic signal is a clean absence: the KT2440 proteome (UP000000556) contains **no S-adenosylmethionine decarboxylase (SpeD, EC 4.1.1.50)**. Because the classical route requires SpeD to generate the decarboxylated-SAM aminopropyl donor consumed by spermidine synthase, the lone spermidine synthase annotation in the genome (PP_1867 / Q88LR2) cannot support de novo spermidine synthesis on its own. Lacking SpeD while still making spermidine is the recognized molecular signature of organisms that run the carboxyspermidine route ([PMID: 22025614](#)).

The two committed enzymes of the carboxyspermidine route are present and, importantly, **genomically adjacent** — the hallmark cluster organization described for this pathway. **PP_2929 (nspC / Q88IS0)** is a bona fide **carboxyspermidine/carboxynorspermidine decarboxylase (CASDC/CANSDC, EC 4.1.1.96)**: it is 66% identical to a reviewed nspC ortholog and only ~28% identical to its own genome's fold-III PLP decarboxylase paralogs, so it is not an over-propagated annotation. Immediately upstream, **PP_2928 (Q88IS1)**, currently mis-annotated as a "saccharopine dehydrogenase," is in fact the **carboxyspermidine dehydrogenase (CASDH, EC 1.5.1.43)** — it belongs to the correct saccharopine-dehydrogenase/homospermidine-synthase family (IPR005097), is the right length (414 aa), sits directly next to nspC, and is 46.8% identical over its full length to the

experimentally characterized *Vibrio cholerae* CANSDH (VC_1624). PP_2928 is not in the supplied candidate list; it is the single most consequential curation action arising from this review.

For module curation, the carboxyspermidine spermidine-biosynthesis module is **satisfiable** in KT2440: putrescine is supplied redundantly (SpeA/AguA/SpeB/SpeC, all in the candidate set), aspartate- β -semialdehyde is supplied by Asd (PP_1989), CASDH is PP_2928, and CASDC is PP_2929. The classical SpeD/SpeE step should be marked **not_expected_in_target_taxon**. Critically, polyamine biosynthesis annotations must **not** be transferred from *P. aeruginosa* to *P. putida*: the two species use opposite routes, making this a concrete anti-over-propagation caveat within a single genus.

2. Target-Organism Pathway Definition

What the process is

"Carboxyspermidine biosynthesis" is the committed, two-enzyme core of the **alternative (SAM-independent) spermidine biosynthetic pathway**. In this route the aminopropyl donor is **L-aspartate- β -semialdehyde (ASA)** rather than decarboxylated S-adenosylmethionine. The core chemistry is:

1. **CASDH (EC 1.5.1.43)** condenses **putrescine + ASA** in an **NADPH-dependent** reductive amination to form **carboxyspermidine** ([PMID: 36283333](#)).
2. **CASDC (EC 4.1.1.96)**, a **PLP-dependent fold-III decarboxylase**, removes the carboxyl group to yield **spermidine**.

When the same two enzymes act on 1,3-diaminopropane instead of putrescine they make sym-norspermidine via carboxynorspermidine — hence the CANSDH/CANSDC nomenclature. The enzymes are substrate-promiscuous; which polyamine is produced in vivo depends on the diamine supplied ([PMID: 22025614](#)).

Boundaries — what to keep separate

- **Putrescine supply** (arginine/agmatine and ornithine decarboxylase branches) feeds this module but is upstream metabolism, largely mapped under KEGG [ppu00330](#) -adjacent arginine metabolism. It should be treated as substrate supply, not part of the committed carboxyspermidine core.

- **Aspartate- β -semialdehyde supply** (Asd, EC 1.2.1.11) is shared with lysine/DAP and threonine/methionine biosynthesis; it is a co-substrate node, not part of the module proper.
- **Classical spermidine synthesis** (SpeD EC 4.1.1.50 + SpeE EC 2.5.1.16) is the *competing* route and should be modeled as a **mutually exclusive alternative**, marked not-expected here.
- The nominal bucket **ppu00330** "**Arginine and proline metabolism**" is an overview map that co-locates dozens of only loosely related reactions (arginine succinyltransferase AST pathway, proline catabolism PutA, 5-oxoprolinases, creatinase, opine oxidase, etc.). Most of the 39 supplied candidates belong to those neighboring processes and are **not** part of carboxyspermidine biosynthesis. The review deliberately narrows to the polyamine sub-network.

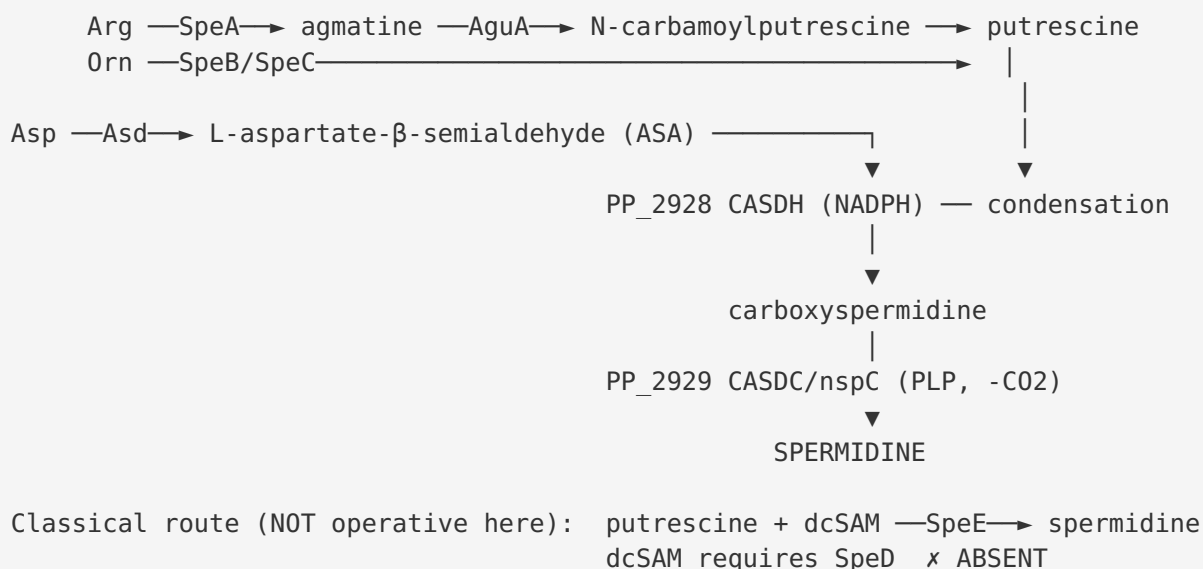
Alternate names / database definitions

- **CASDH** = carboxyspermidine dehydrogenase; **CANS DH** = carboxynorspermidine dehydrogenase (same enzyme family, EC 1.5.1.43).
 - **CASDC/CANS DC** = carboxyspermidine/carboxynorspermidine decarboxylase, gene symbol **nspC**, EC 4.1.1.96.
 - The pathway is variously called the "alternative," "carboxyspermidine," or "aspartate-semialdehyde" spermidine pathway. It is **not** the classical "aminopropyltransferase" route.
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3. Expected Step Model

Step	Reaction	Enzyme (EC)	KT2440 gene	Status
S1	Diamine supply: Arg → agmatine → putrescine, and Orn → putrescine	ADC/ SpeA (4.1.1.19), AguA (3.5.3.12), SpeB (3.5.3.11), SpeC (4.1.1.17)	PP_0567, PP_0266, PP_2196, PP_0864	Covered (candidate list)
S2	Aspartate-β-semialdehyde supply	Asd (1.2.1.11)	PP_1989 (not in candidate list)	Covered (housekeeping)
S3	Putrescine + ASA → carboxyspermidine (NADPH)	CASDH (1.5.1.43)	PP_2928 (mis-annotated "saccharopine dehydrogenase")	Covered after re-annotation
S4	Carboxyspermidine → spermidine (PLP, −CO ₂)	CASDC/nspC (4.1.1.96)	PP_2929	Covered (candidate list)
—	Classical alternative: dcSAM donor generation	SpeD (4.1.1.50)	absent	not_expected_in_target_taxon
—	Classical alternative: putrescine + dcSAM → spermidine	SpeE (2.5.1.16)	PP_1867 present but non-functional without SpeD	Not the operative route

ASCII overview of the operative route:



4. Candidate Genes and Evidence

High-confidence, module-defining genes

PP_2929 — nspC / CASDC (Q88IS0, EC 4.1.1.96) — COVERED, high confidence. This is the decarboxylase that produces spermidine. Global Needleman–Wunsch alignments place it firmly in the CANSDC/CASDC family: **66.2% identity** to reviewed *Herminiimonas arsenicoxydans* nspC (Q5QCP2), **51.0%** to *V. cholerae* nspC (Q9KRL4), and **41.7%** to *Campylobacter jejuni* nspC (A8FNH9). By contrast, identity to the co-encoded fold-III PLP decarboxylase paralogs in KT2440 is only **28.3%** (lysA/DAP decarboxylase, Q88L58) and **29.2%** (speC/ornithine decarboxylase, Q88PI6). The >20-percentage-point margin over the paralogs shows PP_2929 is a genuine CASDC, not an over-propagated fold-III annotation. Evidence type: strong homology to *reviewed/experimentally validated* orthologs; direct KT2440 experiment absent but transfer is **strong**. Curation caveat: because CASDC and CANSDC are the same enzyme, the label should note substrate promiscuity; the *in vivo* product is spermidine when putrescine is the diamine (PMID: 22025614).

PP_2928 — CASDH (Q88IS1, EC 1.5.1.43) — NOT in candidate list; PROMOTE and RE-ANNOTATE. PP_2928 is currently annotated "saccharopine dehydrogenase," which is why it was excluded from the polyamine candidate set. Three independent lines of evidence identify it as the missing CASDH: (i) it carries **IPR005097 (Saccharopine dehydrogenase/**

homospermidine synthase, NADP-dependent domain) plus IPR032095 and the IPR036291 NAD(P)-binding superfamily fold — the exact family to which characterized CASDH/CANS DH proteins belong; (ii) it is **414 aa**, matching CASDH references (*V. cholerae* VC_1624 Q9KRL3 = 414 aa; *V. alginolyticus* PODPE4 = 414 aa); and (iii) it sits **immediately adjacent to PP_2929/nspC**, reproducing the CASDH–CASDC gene cluster reported across bacteria ([PMID: 22025614](#)). A full-length global alignment to *V. cholerae* CANS DH gives **46.8% identity (203/434 columns)** — well above over-propagation noise and consistent with true orthology. Evidence type: strong homology + syntenic context; transfer **strong**. This is the single most important curation action from the review.

Putrescine-supply genes (substrate provision — covered)

Gene	Locus	EC	Role	Note
speA	PP_0567	4.1.1.19	Arginine decarboxylase → agmatine	Covered
aguA	PP_0266	3.5.3.12	Agmatine deiminase → N-carbamoylputrescine	Covered
speB	PP_2196	3.5.3.11	Agmatinase → putrescine	Covered; note a second agmatinase annotation (PP_4523) exists — paralog ambiguity
speC	PP_0864	4.1.1.17	Ornithine decarboxylase → putrescine	Covered (primary bucket <code>ppu04148</code> in metadata)

These provide redundant putrescine supply, so S1 is robustly satisfied regardless of which branch dominates under a given condition.

Candidate-list genes that are NOT part of carboxyspermidine biosynthesis

The great majority of the 39 supplied candidates are `ppu00330` neighbors unrelated to the module and should not be counted toward module satisfiability. Notable examples and their true homes:

- **AST arginine-succinyltransferase pathway** (arginine catabolism): astA-I/PP_4479, astA-II/PP_4480, astB/PP_4477, astD/PP_4478, astE/PP_4475.
- **Proline metabolism**: putA/PP_4947 (Pro→Glu), proI/PP_5095 (P5C reductase), pip/PP_5028 (prolyl aminopeptidase), proA/PP_4811, proB/PP_0691.

- **Guanidino/arginine side reactions:** aruH/PP_3721, aruI/PP_3723, kauB/PP_5278, creA/PP_3667 (creatinase), ocd/PP_3533 (ornithine cyclodeaminase).
- **5-oxoprolinase:** oplA/PP_3515, oplB/PP_3514.
- **Opine oxidase:** ooxA/PP_4456, ooxB/PP_4457.
- **Generic oxidoreductases/aminotransferases with broad EC (annotation risk):** PP_3146, PP_4548, PP_5273, PP_2588 (class III aminotransferase), PP_2589 (aldehyde dehydrogenase family), spuC-II/PP_5182 (polyamine:pyruvate transaminase — polyamine *catabolism/interconversion*, not biosynthesis).
- **Clearly mis-bucketed:** codA/PP_3189 (cytosine deaminase, pyrimidine salvage), PP_4983 (tryptophan 2-monooxygenase), ldcC/PP_4140 (lysine decarboxylase → cadaverine), patD/PP_1481 and prr/PP_2801 (γ -aminobutyraldehyde dehydrogenases, putrescine *catabolism*).

None of these should be marked as covering carboxyspermidine steps.

The non-operative classical route

PP_1867 (Q88LR2) is annotated as a spermidine synthase (SpeE-type, 251 aa, IPR029063). It is present but **cannot** drive de novo spermidine synthesis because its co-substrate, decarboxylated SAM, is not made — there is **no SpeD (EC 4.1.1.50)** anywhere in the proteome. This mirrors the *Helicobacter pylori* precedent, where a spermidine-synthase annotation coexists with an incomplete, non-operative classical pathway ([PMID: 28648602](#)). PP_1867 may function in a salvage/interconversion context but should not be scored as the biosynthetic route.

5. Gaps, Ambiguities, and Likely Over-Annotations

Resolved apparent gap (the key result). At first pass CASDH looks missing because PP_2928 is labeled "saccharopine dehydrogenase." This is a **false gap**: the saccharopine-dehydrogenase and CASDH activities share the same NADP-binding fold family (IPR005097), and the syntenic + length + sequence-identity evidence collectively identify PP_2928 as CASDH. Marking S3 as a gap would be an error; it should be **covered after re-annotation**.

Genus-level over-propagation risk (hard caveat). *P. aeruginosa* PAO1 encodes the **classical** machinery — two SAM decarboxylases (speD/PA0654 Q9I5R7; speH/PA4773 Q9HV35) and two spermidine synthases (speE1/PA1687 Q9X6R0; speE2/PA4774 Q9HV34) — and has **zero** CASDC (EC 4.1.1.96) or CANS DH (EC 1.5.1.43). *P. putida* KT2440 is the inverse: zero SpeD, one SpeE-type

synthase (PP_1867), and a complete PP_2928–PP_2929 carboxyspermidine operon. Automated transfer of polyamine annotations from *P. aeruginosa* to *P. putida* (or vice versa) would be **wrong** despite same-genus proximity. This is a concrete, documented anti-over-propagation example.

Paralog ambiguities to flag. - Two agmatinase annotations (speB/PP_2196 and PP_4523) — resolve which is the physiological agmatinase. - PP_2929 vs. the fold-III decarboxylase paralogs (lysA Q88L58, speC Q88PI6): resolved here by the >20-point identity margin, but worth recording so future automated pipelines do not reassign it. - Multiple "oxidoreductase"/aminotransferase candidates (PP_3146, PP_4548, PP_5273, PP_2588) carry broad EC/GO terms and are prime over-annotation candidates; none belong to this module.

Broad EC/GO mappings. spuC-II/PP_5182 (polyamine:pyruvate transaminase) and the aldehyde-dehydrogenase family proteins are polyamine-*catabolic* and should not inflate biosynthetic module coverage.

6. Module and GO-Curation Recommendations

Module step	Recommended status	Rationale
Putrescine supply (S1)	covered	SpeA/Agua/SpeB/SpeC all present (candidate list)
ASA supply (S2)	covered	Asd/PP_1989 (housekeeping, EC 1.2.1.11)
Carboxyspermidine synthesis — CASDH (S3)	covered — requires re-annotation of PP_2928	family + synteny + 46.8% identity to <i>V. cholerae</i> CANSDH
Carboxyspermidine decarboxylation — CASDC (S4)	covered	PP_2929/nspC, 66% to reviewed nspC
Classical SpeD (dcSAM generation)	not_expected_in_target_taxon	zero EC 4.1.1.50 in proteome
Classical SpeE step	not the operative route	PP_1867 present but non-functional without SpeD

Module-boundary verdict: the carboxyspermidine spermidine-biosynthesis module is **satisfiable** in *P. putida* KT2440. The generic module should explicitly model the classical vs. carboxyspermidine routes as **mutually exclusive alternatives** and record that KT2440 uses the latter.

GO / annotation requests: - Re-annotate **PP_2928** with GO:0034070 / EC 1.5.1.43 activity (carboxyspermidine dehydrogenase / "carboxynorspermidine dehydrogenase") and demote the saccharopine-dehydrogenase label to a fold-level note. - Confirm **PP_2929** annotation as CASDC (EC 4.1.1.96) with the caveat that in vivo product is spermidine. - Add a taxon-scoped rule preventing propagation of classical SpeD/SpeE polyamine annotations into *P. putida*-type genomes lacking SpeD.

No new module document is required — the existing alternative-spermidine module fits — but the KT2440 instance needs the PP_2928 re-annotation and the SpeD not-expected flag to be recorded.

7. Genes to Promote to Full fetch-gene Review

1. **PP_2928 (Q88IS1)** — *highest priority*. Mis-annotated "saccharopine dehydrogenase" that is almost certainly CASDH. Full review should confirm catalytic residues, NADPH dependence, and operon context; this is the linchpin of module satisfiability.
 2. **PP_2929 (Q88IS0 / nspC)** — confirm CASDC assignment and document the CASDC/CANSDC substrate promiscuity for the curated record.
 3. **PP_1867 (Q88LR2)** — clarify the role of the orphan spermidine synthase in a SpeD-null background (salvage? non-functional? interconversion?).
 4. **PP_4523 (Q88EE2) vs PP_2196 (SpeB)** — resolve the duplicate agmatinase annotations feeding putrescine supply.
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8. Key References

- **PMID: 19196710** — Lee et al. *An alternative polyamine biosynthetic pathway is widespread in bacteria and essential for biofilm formation in Vibrio cholerae*. Defines the ASA-dependent route (fused DABA AT/DC + CANSDH + CANSDC) and shows deletion of either CANSDH or CANSDC abolishes both norspermidine and spermidine — the basis for requiring both PP_2928 and PP_2929.
- **PMID: 22025614** — *Alternative spermidine biosynthetic route... dominant polyamine pathway in human gut microbiota*. Establishes that lacking SpeD/spermidine synthase is the signature of carboxyspermidine users, that CASDH/CASDC are clustered with putrescine-biosynthetic genes, and that the nspC ortholog makes spermidine (not norspermidine) in vivo.
- **PMID: 36283333** — *Structural analysis of carboxyspermidine dehydrogenase from Helicobacter pylori*. Defines the CASDH reaction (putrescine + ASA → carboxyspermidine, NADPH-dependent) that PP_2928 is proposed to catalyze.
- **PMID: 28648602** — *Helicobacter pylori does not use spermidine synthase to produce spermidine*. Precedent that a spermidine-synthase annotation can coexist with an incomplete, non-operative classical pathway (parallels PP_1867 in a SpeD-null genome).
- **PMID: 35931895** — Engineered *E. coli* co-expressing HSD + CASDH + CASDC for spermidine production; independent confirmation of the two committed enzyme activities and their NADPH requirement.

- Supporting *P. putida* polyamine context: [PMID: 29886249](#) (*P. putida* modulation of host polyamines), [PMID: 26662997](#) (KT2440 spermidine/putrescine chemoreceptor binding), [PMID: 31451546](#) (arginine biosynthesis in *P. putida*).

Limitations and Knowledge Gaps

- All KT2440-specific assignments rest on **sequence homology, InterPro domains, protein length, and genomic syntenicity** — there is **no direct biochemical or genetic experiment in KT2440** confirming CASDH activity for PP_2928 or CASDC activity for PP_2929. Transfer from *V. cholerae*, *H. pylori*, and reviewed nspC orthologs is judged strong but remains inferential.
- The physiological product (spermidine vs. sym-norspermidine) has not been measured in KT2440; the norspermidine branch appears absent, but this was inferred, not assayed.
- The role of the orphan spermidine synthase PP_1867 in a SpeD-null background is unresolved.
- Putrescine-supply redundancy (two agmatinases) was noted but the physiologically dominant route was not established.

Proposed Follow-up Experiments / Actions

1. **Re-annotate PP_2928 as CASDH** in the curated record (highest-value single action) and promote it to full `fetch-gene` review.
 2. **Targeted deletions** of PP_2928 and PP_2929 in KT2440 with LC-MS polyamine profiling — the definitive test (mirrors the *V. cholerae* knockouts, [PMID: 19196710](#)).
 3. **In vitro assays**: express PP_2928 and test NADPH-dependent condensation of putrescine + ASA → carboxyspermidine; express PP_2929 and test carboxyspermidine → spermidine.
 4. **Metabolite confirmation**: quantify spermidine and check for sym-norspermidine to confirm the operative diamine input.
 5. **Add a taxon-scoped propagation guard** preventing classical SpeD/SpeE polyamine annotations from being transferred into SpeD-null *P. putida*-type genomes.
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